

A Planar Equality Case Outside the Source-Style Pattern for the Dual Log-Brunn-Minkowski Inequality

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Source and Target

The source paper is Christos Saroglou, *More on logarithmic sums of convex bodies*, arXiv:1409.4346 [1]. In Section 6, Saroglou proves the dual logarithmic Brunn-Minkowski inequality

$$|(\lambda \cdot K +_0 (1 - \lambda) \cdot L)^\circ| \leq |K^\circ|^\lambda |L^\circ|^{1-\lambda}, \quad 0 < \lambda < 1.$$

Immediately after the theorem, the paper says that it is plausible that the equality cases are the same as those in Conjecture 1.1, with non-symmetric bodies allowed, but does not address this.

equality (see Corollary 6.5 below).

Theorem 6.1. *Let K, L be two convex bodies in $\mathbb{R}^n$ and $\lambda \in (0, 1)$. Then,*

$$|(\lambda \cdot K +_0 (1 - \lambda) \cdot L)^\circ| \leq |K^\circ|^\lambda |L^\circ|^{1-\lambda}. \quad (11)$$

Once Theorem 6.1 is established, one can follow Firey's argument [14] to prove Corollary 6.5 (see below), where the volume is replaced by the other quermassintegrals. Since the dual L^0 -sum contains the dual L^p -sum, for $p \geq 0$, Theorem 6.1 extends immediately to the L^p -setting, for all $p \geq 0$. Therefore, it is stronger than Firey's [13] dual Brunn-Minkowski inequality. It is also stronger than the dual Brunn-Minkowski inequality with respect to L^0 -radial sums, established in [20]. It seems plausible that the equality cases in (11) are exactly the equality cases in Conjecture 1.1 (here of course non-symmetric bodies are allowed); we do not address this here.

Without loss of generality, we may assume that K and L contain 0 in their interiors. Otherwise, the assertion would be trivial. As in the previous section, we will prove our claim for the (asymmetric) discrete approximations of the logarithmic sum K and L . The rest of the proof will follow by compactness. Set

$$AR_\lambda = \{x \in \mathbb{R}^n \mid x \cdot v_i \leq r_i^\lambda s_i^{1-\lambda}, \ i = 1, \dots, m\},$$

where $v_1, \dots, v_m \in S^{n-1}$, $r_i = h_K(v_i)$, $s_i = h_L(v_i)$, $i = 1, \dots, m$. Since K and L contain 0 in their interiors, it is true that $r_i, s_i > 0$, $i = 1, \dots, m$, thus AR_λ is well defined. One, then, needs to prove that the function $(0, 1) \ni \lambda \mapsto |(AR_\lambda)^\circ|$ is log-convex (i.e. its logarithm is convex). On the other hand.

Figure 1: Theorem 6.1 and the equality-case comment in arXiv:1409.4346, p. 17.

The construction below gives equality in the dual inequality for a planar pair which is neither homothetic nor of the natural product/dilate form suggested by the equality cases in Conjecture 1.1. Thus the source-style equality-case analogy is false in this literal sense.

Definitions

For a convex body $C \subset \mathbb{R}^2$ containing the origin in its interior, write

$$C^\circ = \{y \in \mathbb{R}^2 : x \cdot y \leq 1 \text{ for all } x \in C\}$$

for its polar. The logarithmic sum is the Wulff body determined by the support-function inequality

$$\lambda \cdot K +_0 (1 - \lambda) \cdot L = \{y \in \mathbb{R}^2 : y \cdot u \leq h_K(u)^\lambda h_L(u)^{1-\lambda} \text{ for all } u \in \mathbb{R}^2\}.$$

Proof Intuition

Saroglou's proof of the dual inequality reduces the volume estimate to log-convexity of the volume of polytopes whose vertices move by independent exponential radial factors. In the plane, the area of a polygon is the sum of determinants of adjacent vertices. If the radial exponents alternate 0, 1, 0, 1 around a quadrilateral, every adjacent determinant has exponent 1. The area is therefore exactly log-affine.

Taking polars turns this log-affine moving quadrilateral into a pair K, L . The logarithmic sum has polar containing the intermediate moving quadrilateral, while the dual log-Brunn-Minkowski inequality gives the opposite volume bound. Hence equality is forced for the true logarithmic sum, not merely for a discrete approximation. The chosen quadrilateral is deliberately not a product polar, so this equality falls outside the natural source-style product/dilate list.

The Counterexample

Theorem 1. *Let*

$$x_1 = (2, 0), \quad x_2 = (0, 1), \quad x_3 = (-1, \tfrac{1}{2}), \quad x_4 = (0, -1),$$

and, for $0 \leq t \leq 1$, set

$$P_t = \text{conv}\{x_1, e^t x_2, x_3, e^t x_4\}.$$

Put $K = P_0^\circ$ and $L = P_1^\circ$. Then, for every $0 < \lambda < 1$,

$$|(\lambda \cdot K +_0 (1 - \lambda) \cdot L)^\circ| = |K^\circ|^\lambda |L^\circ|^{1-\lambda}.$$

Nevertheless, K is not a two-dimensional Cartesian product of one dimensional factors, even after a linear change of coordinates, and L is not a dilate of K . Hence this equality pair is outside the natural planar product/dilate equality pattern suggested by the source comment.

Proof. First note that the four listed points occur in counterclockwise order and that P_t contains the origin in its interior for $0 \leq t \leq 1$. Indeed, with $q = e^t \geq 1$, direct edge-orientation checks give

$$2q, \quad q, \quad q, \quad 2q$$

for the signed areas of the triangles formed by the origin and consecutive vertices, and the edge-cross-products of the polygon are

$$3q - 1, \quad 2q, \quad 3q + 1, \quad 4q,$$

all positive.

The area of P_t is especially simple. By the shoelace formula,

$$\begin{aligned} 2|P_t| &= \det(x_1, qx_2) + \det(qx_2, x_3) + \det(x_3, qx_4) + \det(qx_4, x_1) \\ &= 2q + q + q + 2q = 6q. \end{aligned}$$

Thus

$$|P_t| = 3e^t. \tag{1}$$

Let $a_1 = a_3 = 0$ and $a_2 = a_4 = 1$, so that

$$P_t = \text{conv}\{e^{a_i t} x_i : 1 \leq i \leq 4\}.$$

Since each x_i is an exposed vertex of $P_0 = K^\circ$, its polar constraint is active and

$$h_K(x_i) = 1.$$

Likewise $e^{a_i} x_i$ is the corresponding exposed vertex of $P_1 = L^\circ$, so

$$h_L(x_i) = e^{-a_i}.$$

Fix $0 < \lambda < 1$ and write

$$M_\lambda = \lambda \cdot K +_0 (1 - \lambda) \cdot L.$$

The defining support inequalities for the logarithmic sum, tested only in the four directions x_i , imply

$$M_\lambda \subseteq Q_\lambda := \{y \in \mathbb{R}^2 : y \cdot x_i \leq e^{-(1-\lambda)a_i}, 1 \leq i \leq 4\}.$$

Taking polars reverses inclusion, and the polar of Q_λ is

$$Q_\lambda^\circ = \text{conv}\{e^{(1-\lambda)a_i} x_i : 1 \leq i \leq 4\} = P_{1-\lambda}.$$

Hence

$$P_{1-\lambda} \subseteq M_\lambda^\circ. \tag{2}$$

Saroglou's dual logarithmic Brunn-Minkowski inequality gives

$$|M_\lambda^\circ| \leq |K^\circ|^\lambda |L^\circ|^{1-\lambda} = |P_0|^\lambda |P_1|^{1-\lambda}.$$

Using (1),

$$|P_0|^\lambda |P_1|^{1-\lambda} = 3^\lambda (3e)^{1-\lambda} = 3e^{1-\lambda} = |P_{1-\lambda}|.$$

Together with (2), this forces equality:

$$|M_\lambda^\circ| = |P_{1-\lambda}| = |K^\circ|^\lambda |L^\circ|^{1-\lambda}.$$

It remains to verify that this is not a source-style planar equality case. In dimension two, a Cartesian product of one-dimensional factors, after a linear change of coordinates, has a polar whose vertices split into two pairs lying on two lines through the origin. The vertices of $P_0 = K^\circ$ do not admit such a split. The only origin-collinear pair among the four vertices is $\{x_2, x_4\}$, while

$$\det(x_1, x_3) = 1 \neq 0.$$

Thus K is not a two-dimensional product body of one-dimensional factors.

Finally, L is not a dilate of K . If $L = cK$, then

$$P_1 = L^\circ = c^{-1}K^\circ = c^{-1}P_0.$$

But P_1 has the two y -axis vertices ex_2, ex_4 , whereas P_0 has x_2, x_4 , so any scalar relation would force $c^{-1} = e$. The same scalar would then have to send x_1 to ex_1 , but x_1 itself is a vertex of P_1 . This contradiction proves that L is not a dilate of K . $\square$

Verification Notes

The script `code/check_quadrilateral_equality.py` verifies the exact determinants

$$2, 1, 1, 2$$

which give $|P_t| = 3e^t$, checks that P_0 has only one origin-collinear pair of vertices, and records the elementary obstruction to a scalar homothety $P_1 = cP_0$.

Command used:

```
conda run --no-capture-output -n sandbox python
code/check_quadrilateral_equality.py
```

This computation is not a substitute for the proof; it is a reproducible sanity check for the coordinates used in the construction.

Scope and Novelty Check

This packet does not refute Saroglou’s dual log-Brunn-Minkowski inequality. It refutes the equality-case analogy if that analogy is read as requiring the same planar product/dilate pattern as in the log-Brunn-Minkowski equality statement. It also does not classify all equality cases in the dual theorem.

The run’s cheap indexes were searched for arXiv:1409.4346, the source title, “log-Brunn”, “dual log-Brunn-Minkowski”, “equality cases”, “B-conjecture”, “cone-volume”, and related log-Minkowski terms. No prior packet for this paper or route was found.

A bounded web search on 2026-07-18 for exact and nearby phrases including “dual log-Brunn-Minkowski equality cases”, “Saroglou dual B-conjecture equality cases polytopes”, and the source title found the source article and later general log-Brunn-Minkowski literature, but no existing equality-case classification or counterexample for this Section 6 comment.

Human Review Recommendation

Recommended verdict: likely valid counterexample to the equality-case analogy, with narrow scope. Reviewers should mainly check the intended interpretation of the source’s informal equality-case comment: the mathematical equality above is exact, while the classification conclusion being refuted is the natural source-style product/dilate pattern rather than a formal theorem.

References

- [1] Christos Saroglou, *More on logarithmic sums of convex bodies*, arXiv:1409.4346; *Mathematika* 62 (2016), 818–841.